

Basic Programs

1. Print "Welcome to Java Programming"

```
class Welcome {  
  
    public static void main(String[] args) {  
  
        System.out.println("Welcome to Java Programming");  
  
    }  
  
}
```

2. Using print() and println()

```
class PrintDemo {  
  
    public static void main(String[] args) {  
  
        System.out.print("Hello ");  
  
        System.out.println("Java");  
  
    }  
  
}
```

3. Simple Interest

```
class SimpleInterest {  
  
    public static void main(String[] args) {  
  
        double p = 1000, r = 5, t = 2;  
  
        double si = (p * r * t) / 100;  
  
        System.out.println("Simple Interest = " + si);  
  
    }  
  
}
```

4. Area of a Square

```
class SquareArea {  
  
    public static void main(String[] args) {  
  
        int side = 4;  
  
        System.out.println("Area = " + (side * side));  
  
    }  
  
}
```

5. Area of a Circle

```
class CircleArea {  
  
    public static void main(String[] args) {  
  
        double r = 7;  
  
        System.out.println("Area = " + (Math.PI * r * r));  
  
    }  
  
}
```

6. Swap using temp variable

```
class SwapTemp {  
  
    public static void main(String[] args) {  
  
        int a = 10, b = 20, temp;  
  
        temp = a;  
  
        a = b;  
  
        b = temp;  
  
        System.out.println(a + " " + b);  
  
    }  
  
}
```

```
}
```

7. Swap without temp

```
class SwapNoTemp {  
  
    public static void main(String[] args) {  
  
        int a = 10, b = 20;  
  
        a = a + b;  
  
        b = a - b;  
  
        a = a - b;  
  
        System.out.println(a + " " + b);  
  
    }  
}
```

8. BMI Calculation

```
class BMI {  
  
    public static void main(String[] args) {  
  
        double weight = 60;  
  
        double height = 1.7;  
  
        double bmi = weight / (height * height);  
  
        System.out.println("BMI = " + bmi);  
  
    }  
}
```

9. Product without * operator

```
class Multiply {
```

```
public static void main(String[] args) {  
    int a = 5, b = 4, result = 0;  
    for(int i = 1; i <= b; i++)  
        result += a;  
    System.out.println(result);  
}  
}
```

10. 2 to the power 10

```
class Power {  
    public static void main(String[] args) {  
        System.out.println(Math.pow(2, 10));  
    }  
}
```

11. Default values of data types

```
class DefaultValues {  
    static int i;  
    static float f;  
    static double d;  
    static char c;  
    static boolean b;  
    static String s;  
  
    public static void main(String[] args) {
```

```
        System.out.println(i);

        System.out.println(f);

        System.out.println(d);

        System.out.println(c);

        System.out.println(b);

        System.out.println(s);

    }

}
```

12. Even/Odd using ternary

```
class EvenOdd {

    public static void main(String[] args) {

        int n = 7;

        System.out.println(n % 2 == 0 ? "Even" : "Odd");

    }

}
```

Ternary Operator

Passed / Failed

```
class Result {

    public static void main(String[] args) {

        int mark = 45;

        System.out.println(mark >= 40 ? "Passed" : "Failed");

    }

}
```

Decision Control Constructs

13. Divisible by 7

```
class Div7 {  
  
    public static void main(String[] args) {  
  
        int n = 49;  
  
        System.out.println(n % 7 == 0);  
  
    }  
}
```

14. Even/Odd

```
class EvenOddIf {  
  
    public static void main(String[] args) {  
  
        int n = 10;  
  
        if(n % 2 == 0) System.out.println("Even");  
  
        else System.out.println("Odd");  
  
    }  
}
```

15. +ve, -ve or zero

```
class NumberType {  
  
    public static void main(String[] args) {  
  
        int n = -5;  
  
        if(n > 0) System.out.println("Positive");  
  
        else if(n < 0) System.out.println("Negative");  
  
    }  
}
```

```
        else System.out.println("Zero");
    }
}
```

16. Day using if-else

```
class Day {
    public static void main(String[] args) {
        int d = 3;
        if(d == 1) System.out.println("Monday");
        else if(d == 2) System.out.println("Tuesday");
        else if(d == 3) System.out.println("Wednesday");
        else System.out.println("Invalid");
    }
}
```

17. Month using switch

```
class Month {
    public static void main(String[] args) {
        int m = 1;
        switch(m) {
            case 1: System.out.println("January"); break;
            case 2: System.out.println("February"); break;
            default: System.out.println("Invalid");
        }
    }
}
```

```
}
```

18. Leap Year

```
class LeapYear {  
  
    public static void main(String[] args) {  
  
        int y = 2024;  
  
        System.out.println((y%4==0 && y%100!=0) || y%400==0);  
  
    }  
}
```

19. Max of 3 numbers

```
class MaxThree {  
  
    public static void main(String[] args) {  
  
        int a=10,b=20,c=15;  
  
        int max = Math.max(a, Math.max(b, c));  
  
        System.out.println(max);  
  
    }  
}
```

20. Greatest & Least

```
class GreatLeast {  
  
    public static void main(String[] args) {  
  
        int a=10,b=20,c=5;  
  
        System.out.println(Math.max(a, Math.max(b,c)));  
  
        System.out.println(Math.min(a, Math.min(b,c)));  
  
    }  
}
```



```
}  
}
```

Looping Control Constructs

21. Name 10 times

```
class Name {  
    public static void main(String[] args) {  
        for(int i=1;i<=10;i++)  
            System.out.println("Your Name");  
    }  
}
```

22. 1–20

```
class OneToTwenty {  
    public static void main(String[] args) {  
        for(int i=1;i<=20;i++)  
            System.out.println(i);  
    }  
}
```

23. 10–1

```
class TenToOne {  
    public static void main(String[] args) {  
        for(int i=10;i>=1;i--)  
            System.out.println(i);  
    }  
}
```

```
}  
}
```

24. Sum of n natural numbers

```
class SumN {  
  
    public static void main(String[] args) {  
  
        int n = 10, sum = 0;  
  
        for(int i=1;i<=n;i++)  
  
            sum += i;  
  
        System.out.println(sum);  
  
    }  
}
```

25. While loop 1–10

```
class WhileLoop {  
  
    public static void main(String[] args) {  
  
        int i=1;  
  
        while(i<=10)  
  
            System.out.println(i++);  
  
    }  
}
```

26. Factorial

```
class Factorial {  
  
    public static void main(String[] args) {
```

```
int n=5, fact=1;

for(int i=1;i<=n;i++)

    fact*=i;

System.out.println(fact);

}

}
```

27. Simple Interest (1–10%)

```
class SIAllRates {

    public static void main(String[] args) {

        int p=1000,t=1;

        for(int r=1;r<=10;r++)

            System.out.println((p*r*t)/100);

    }

}
```

28. Skip 7 & 8

```
class Skip {

    public static void main(String[] args) {

        for(int i=1;i<=10;i++) {

            if(i==7 || i==8) continue;

            System.out.println(i);

        }

    }

}
```

29. Break at 5

```
class BreakDemo {  
  
    public static void main(String[] args) {  
  
        for(int i=1;i<=10;i++) {  
  
            if(i==6) break;  
  
            System.out.println(i);  
  
        }  
  
    }  
  
}
```

30. Pattern

```
class Pattern {  
  
    public static void main(String[] args) {  
  
        for(int i=1;i<=3;i++) {  
  
            for(int j=1;j<=3;j++) {  
  
                if(i==2 && j!=1) continue;  
  
                System.out.println(i+" "+j);  
  
            }  
  
        }  
  
    }  
  
}
```